

Skills Summary

Using Dilations to Reason About Similarity

Use this reference while completing your worksheet or when studying.

Skill 1 — Coordinates of a dilated point

Rules:

Centre at the origin: $(x, y) \mapsto (kx, ky)$.

Centre at (h, m) : $(x, y) \mapsto (h + k(x - h), m + k(y - m))$ — measure from the centre, multiply by k , step back out.

$k > 1$ enlarges, $0 < k < 1$ reduces, and $k = 1$ leaves the figure alone.

Example: A dilation about the origin with $k = 3$ maps $P(4, -2)$ to P' . Find the x -coordinate of P' .

Step 1. The centre is the origin, so each coordinate is just multiplied by k ; take the x -coordinate on its own.

Step 2. $3 \times 4 = 12$ (the y -coordinate is handled the same way).

Try it: A dilation about the origin with $k = 0.5$ maps $Q(5, -4)$ to Q' . Find the y -coordinate of Q' .

check: 2

Skill 2 — Scale factor from two lengths

Rule:

$$k = \frac{\text{image length}}{\text{original length}}$$

Image over original, in that order: k answers “how many times as long is the image?” Lengths on a grid come from the distance formula.

Example: A model edge is 12 cm long. On the enlarged copy the matching edge is 18 cm. Find k .

Step 1. The two edges correspond, so divide the image length by the original length.

Step 2. $18 \div 12 = 1.5$, so every length on the copy is 1.5 times the model's.

Try it: An 8 cm edge becomes 20 cm on the enlarged copy. Find k .

check: 2.5

Skill 3 — Slope under a dilation

Rule:

A dilation maps a segment to a **parallel** segment, so the image has the *same slope*. Equal slopes are the coordinate evidence that the angles did not change — which is why the figure and its image are similar, not just the same size.

Example: Find the slope of the segment from $(1, 2)$ to $(3, 6)$. Its image under $k = 4$ runs from $(4, 8)$ to $(12, 24)$.

Step 1. Slope is rise over run, and the claim to test is that the image gives the same number.

Step 2. $(6 - 2)/(3 - 1) = 4/2 = \mathbf{2}$, and the image gives $(24 - 8)/(12 - 4) = 16/8 = 2$ as well.

Try it: Find the slope of the segment from $(2, 1)$ to $(8, 4)$.

check: 2.0

Skill 4 — Similar figures: sides and area

Rules:

In similar figures every pair of matching sides has the *same* ratio k , so a missing side is $k \times$ (matching side).

Lengths scale by k , but area scales by k^2 : both dimensions grew, so the area grew twice.

Example: A triangle has sides 5 cm and 7 cm. A similar triangle has 15 cm matching the 5 cm side. Find the side matching 7 cm.

Step 1. Matching sides share one ratio, so find k from the pair that is fully known, then apply it to the other side.

Step 2. $k = 15 \div 5 = 3$, so the missing side is $3 \times 7 = \mathbf{21}$ cm.

Try it: A triangle has sides 4 cm and 9 cm. A similar triangle has 10 cm matching the 4 cm side. Find the side matching 9 cm (in cm).

check: 22.5

Example: A figure has area 15 square units. It is dilated with $k = 2$. Find the area of the image.

Step 1. Area is a product of two lengths and each one is multiplied by k , so multiply the area by k^2 , not by k .

Step 2. $15 \times 2^2 = 15 \times 4 = \mathbf{60}$ square units.

Try it: A figure has area 24 square units and is dilated with $k = 3$. Find the area of the image (in square units).

check: 216